

Laura Battaglia

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in LinkedIn

🌐 E.U. Citizen

Machine learning and statistics researcher. Expertise in model misspecification, Bayesian machine learning, and generative modelling. Industry background in risk analysis, financial modelling, and banking supervision.

Education

- 2022 – Now  **Ph.D., University of Oxford**, Statistics and Machine Learning
Research interests: *variational inference and MCMC methods, diffusion and flow-based models, frameworks for model misspecification*
Supervisor: *Geoff Nicholls*
- 2019 – 2020  **MSc, Barcelona School of Economics (Pompeu Fabra)**, Data Science (Grade: 9.25/10)
Main courses: *statistical learning, machine learning theory, optimisation, reinforcement learning, natural language processing, data warehousing, network analysis*
Thesis: *Tracking the Economy Using FOMC Speech Transcripts* under supervision of *Omiros Papaspiliopoulos*.
- 2013 – 2016  **MSc, Bocconi University**, Economics (Grade: 107/110)
Main courses: *statistics, mathematics, econometrics, microeconometrics, development economics, political economics, monetary policy*
Thesis: *Evaluating the accuracy of Regression Discontinuity design: an empirical example using experimental data*
- 2013  **Exchange Program, Cornell University**, Finance (GPA: 4.05/4.0)
- 2010 – 2013  **BSc, Bocconi University**, Finance (Grade: 110/110 cum laude)
Thesis: *The endowment effect as an expectations-based effect: an empirical analysis*.

Employment History

- 2023 – Now  **ML Consultant, Italian Association for Financial Analysis**
Focus: *Advisor on machine learning applications in finance; lead of the monthly Finance & AI Monitor*.
- 2020 – 2022  **Data Science and ML research assistant, Barcelona School of Economics**
Key areas: *natural language processing, MCMC and variational methods for high-dimensional inference*
Lecturer in computational ML and programming
Supervised by Prof. Stephen Hansen (University College London) via ERC Grant
- 2016 – 2019  **Supervision Analyst, European Central Bank - Single Supervisory Mechanism**
Expertise: *business model risk, governance, liquidity risk, cooperative banks, crisis management & failing procedures*
Selected in the first SSM Traineeship Programme, then awarded a fixed position in the official analyst recruitment campaign
- 2014  **Consulting Intern, Promontory Financial Group**
Focus: *contributing to a Section 166 Skilled Person Review of customers in financial difficulty of a large UK bank*

Technical Skills

-  **Programming languages:** Python, R, SQL, Bash
-  **Frameworks and libraries:** JAX, PyTorch, TensorFlow, Numpyro, Pandas

Technical Skills (continued)

- **Cloud platforms and tools:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Tableau, LaTeX

Research Articles

- 1 **L. Battaglia**, T. Farghly, S. Cortinovis, D. de Vassimon Manela, A. Shidani, and C. Williams, “Copula diffusion modelling under marginal constraints,” in *Eighth Workshop on Tractable Probabilistic Modeling (UAI)*, 2025.
- 2 **L. Battaglia** and G. Nicholls, “Amortising variational bayesian inference over prior hyperparameters with a normalising flow,” *under review*, 2025.
- 3 D. d. V. Manela*, **L. Battaglia*** (co-first author), and R. J. Evans, “Marginal causal flows for validation and inference,” *Advances in Neural Information Processing Systems*, 2024.
- 4 **L. Battaglia**, T. Christensen, S. Hansen, and S. Sacher, “Inference for regression with variables generated from unstructured data,” *under review*, 2024.
- 5 J. Jewson, L. Li, **L. Battaglia**, S. Hansen, D. Rossell, and P. Zwiernik, “Graphical model inference with external network data,” *Biometrics*, 2024.

Presentations

- **Inference for regression with variables generated by AI or ML**
July 2025, Seminars of the Economics Department @ University of Pisa, Pisa, Italy *Invited Talk*
- **Martingale correction for mean field variational inference**
June 2025, Workshop on Statistics and experimental Design with ML @ INI, Cambridge, UK *Poster*
May 2025, Post-Bayes workshop @ UCL, London, UK *Poster*
- **Marginal causal flows for validation and inference**
December 2024, Conference on Neural Information Processing Systems, Vancouver, Canada *Poster*
December 2024, NeurIPS @ Cambridge, Cambridge, UK *Poster*
- **The Variational Meta-Posterior for the learning rate and prior hyperparameters**
July 2024, ISBA, Venice, Italy *Poster*
March 2023, BayesComp, Levi, Finland *Poster*
- **Network-adjusted graphical modelling**
June 2022, CESifo Venice Summer Institute, Venice, Italy *Poster*

Scholarships & Awards

- 2022 – 2026 ■ **Clarendon Scholarship** (merit scholarship fully funding Oxford PhD program)

Teaching and grading

- **SB2b Statistical Machine Learning**, University of Oxford
Hilary (Winter) 2024
- **SB3.1 Applied Probability**, University of Oxford
Hilary (Winter) 2023
- **SC4 Advanced Topics in Statistical Machine Learning**, University of Oxford
Hilary (Winter) 2022

Teaching and grading (continued)

- **Summer School in Foundations of Data Science**, Barcelona School of Economics
Spring 2021
- **21DM005 Computational Machine Learning II**, Barcelona School of Economics
Spring 2021

Professional service

- 2024 – Now
- **Equality, Diversity, and Inclusion Committee Member**, Dept. of Statistics, Oxford
 - **Senior Advisor of the Young Statisticians Seminar**, Dept. of Statistics, Oxford